



Missing data imputation framework for bridge structural health monitoring based on slim generative adversarial networks

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ABSTRACT

In structural health monitoring (SHM) systems, sensors are important to collect structural responses to assess the load-resistant capacity and health status of structures. However, data loss of these sensors is sometimes inevitable due to communication outages and malfunction, which will result in incorrect diagnosis of structural health status. Conventional lost data imputation methods fundamentally have low implementing efficiency owing to incompact neurons network structures. This paper proposed a slim generative adversarial imputation network (SGAIN) to recover the missing deflection data for bridge SHM systems. This framework uses slim neural networks with a generator-discriminator architecture to capture the valuable information from the non-missing parts of fault sensor and other normal sensors. Based on analysis on the long-term measured deflection data under the ambient and dynamic excitation of a highway-railway dual-purpose bridge, the proposed method shows the efficient and accurate imputation performance under the input scenarios with different correlation levels. The comparative results show the proposed SGAIN significantly outperforms the conventional GAIN model for all three scenarios with different missing rates. The execution velocity of SGAIN model also outperforms GAIN with 10–20% increment for three type scenarios. Such improvement is significantly valuable and it is believed that SGAIN could also have a good imputation capability when transferred to other valuable scenarios.

1. Introduction

To ensure the safety and stability of structures, Structural Health Monitoring (SHM) systems are widely embedded in large infrastructures (e.g. skyscrapers, bridges, and dams) [1–4]. In the last few decades, SHM systems have been upgraded continuously and measured numerous data day and night with the flourishing development of sensor-based monitoring technologies [5–9]. However, data loss, which is caused by transmitting interference, communication outages, sensor faults, etc., remains a major challenge in SHM. Considering the economy and practicality of the SHM systems, a limited number of sensors can only be deployed to monitor key components of the structures. When data missing occurs, the evaluation of the safety and condition of structures can hardly be performed. Therefore, it is significantly desirable to impute the missing data accurately and efficiently in SHM systems.

Recently, many scholars have vigorously explored incomplete data

imputation technology, and the missing data reconstruction method has transformed from model-driven to data-driven methods. At present, since it is difficult to select a suitable and accurate model to meet the requirement of scientific research and engineering practice, there is little systematic research on the traditional model-driven reconstruction method. However, data-driven reconstruction technologies, including statistical analysis, machine learning, and deep learning, become popular and abundant during the development of artificial intelligence.

Over the past years, statistical analysis has been conducted on missing data imputation. As a method for parameter estimation of random missing data, the mean imputation method [10] and multiple imputation method [11] were proposed initially. Later, Hien et al. [12] reported that Maximum Likelihood and least-squares estimation with polynomial equations reconstructed missing data on concrete bridges. It is found that statistical analysis has the advantages of simplicity, but restricted assumptions on data distribution makes the reconstructed

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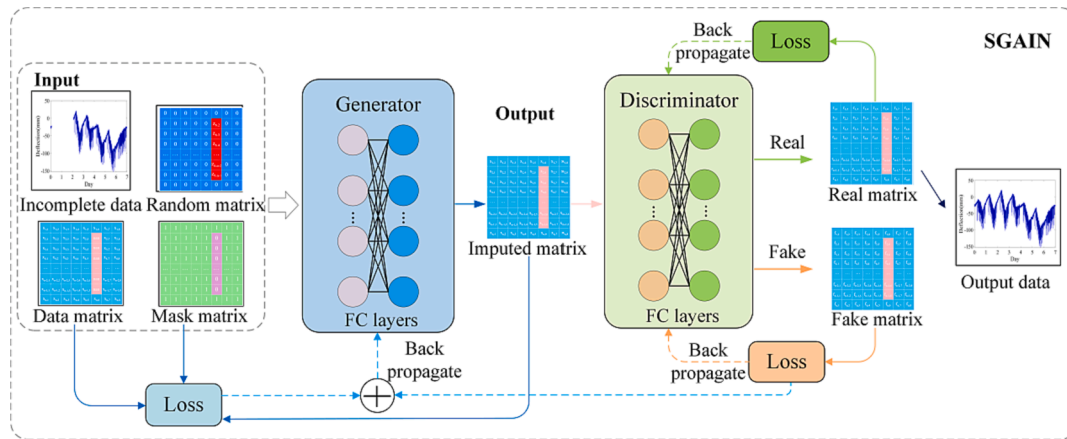


Fig. 1. Detailed configuration of SGAIN.

results poor [13].

There have been several attempts to solve the data reconstruction problems with machine learning and deep learning techniques. In general, data reconstruction can be regarded as a regression application. Ni et al. [14] studied Back-Propagation Neural Network (BPNN) and generalized Regression Neural Network (GRNN) for wind speed and wind pressure reconstruction on Guangzhou Tower. Li et al. [15] demonstrated multi-Task Gaussian Process Regression (mGPR) by imputing the beginning and end of missing temperature and crack displacement. A more accurate model for reconstructing the missing data is called Bayesian Dynamic Linear Models (BDLM) method [16]. The combination reconstruction method can achieve stunning results by taking the advantage of various approaches. Chen et al. [17] developed the Log-Quantile-Density (LQD) and Reproduced Kernel Hilbert Space (RKHS) method to recover the strain data of concrete bridges. Ren [18] proposed Bayesian and Tensor analysis to reconstruct the random missing strain and temperature of concrete bridges, where one-dimensional data were constructed as second or third-order tensors and reliable underlying characteristics were mined rather than directly using original incomplete data. Afterward, Li et al. [19] combined Long Short-Term Memory (Long Short-Term Memory) with empirical mode decomposition (EMD) technique to reconstruct the acceleration responses of cable-stayed bridges. In particular, the decomposition by EMD can improve the accuracy of the core LSTM prediction model. Fan et al. [20] found that Densenets had four architectures: skip connection, dense block, sub-pixel shuffling and drop out, and were utilized to extract complex features and solve the overfitting problem in the acceleration response imputation of Guangzhou Tower. The reconstructed information in the time domain and frequency domain is different based on different criteria for the structural condition evaluation. Zhou et al. [21] performed the Streaming L1-homotopy Algorithm to recover signals for damage identification in the frequency domain. However, such studies are still limited to dealing only with strongly correlated data or large amounts of data.

Compressive Sensing (CS) is a novel signal processing theory emerging in the field of signal processing, which discards the partial amounts of data to mitigate the data sampling and transmission burden. Many researchers also studied the effect of missing data recovery using CS. It was found it is sufficient to reconstruct temporal signals by capturing little information based on sampling theorem and sparsity assumptions [22,23]. Another kind of method was developed on Compressive Sensing that conducts the response reconstruction with non-uniform data loss mode [24]. Moreover, Yang and Nagarajaish [25] applied Sparse representation versus low-rank structure to recover the acceleration data of Tianjin Yonghe concrete Bridge and Guangzhou Tower. However, with such methods, the key information may be ignored in sampling and transmission, which improves the efficiency of

reconstruction but sacrifices the reconstruction accuracy.

In addition, the missing data imputation methods are generally divided into two categories according to whether they are the generative model and the discriminative model. The differences between the two models are that the generative model predicts the conditional probability of missing data through Bayesian theory, while the discriminative model determines the decision boundaries between classes by examining conditional probability distribution. Specifically, the representative generative models include Expectation-Maximization (EM) [26], Denoising Autoencoder (DAE) [27], and multiple imputations by Chained Equations (MICE) [28]. While the discriminative model based methods predominantly include MissForest and Matrix Completion [29]. The advantage of DAE is sufficiently embodied after training the complete data, while the reconstruction effect of DAE is unremarkable for the incomplete dataset. Similarly, the complete response variable is essential to train MissForest model but is necessarily unsatisfied in any incomplete dataset. These researches reported here consistently pointed toward poor computational results and low training efficiency.

Generative adversarial network (GAN) is a state-of-the-art neural network, taking the advantage of both generative model and discriminative model. In follow-up studies, GAN has been continuously improved since it was proposed by Goodfellow [30], including CGAN, WGAN, WGAN-GP, etc. [31,32]. A large number of researchers have conducted extensive research based on GAN, and achieved great success in the fields of image synthesis, image conversion, and audio synthesis, etc. [33]. Moreover, the research to date has been carried out missing data imputation using GAN. Jiang and Hou [34,35] conducted exploratory research to reconstruct the continuous missing strain data with spatiotemporal correlation based on GAN. What we know about SegGAN is largely based on qualitative studies that investigate how to reconstruct the short-term acceleration data in simulation and experiment. The comparative results showed that SegGAN has superior reconstructed performance but with long execution time [36].

Although the present studies have addressed a contemporary light on the critical issue of missing data recovery, the long-term continuous structural response recovery under multichannel spatiotemporal information is still challenging. In SHM systems, deflection is an important identification to assess the load-carrying capacity and health status of bridges. At present, the recovery methods are mainly popular for reconstructing the structural responses such as acceleration, strain, etc., but other valuable scenarios (deflection) to recover missing data with high loss rates are not sufficiently studied. In addition, a key factor that hinders the recovery efficiency and accuracy is that networks don't have a compact and deep structure, while a suitable network structure is the guarantee of quality and efficiency. Moreover, a comparative analysis of recovery mechanisms with and without consideration of structural spatiotemporal information is not implemented yet.

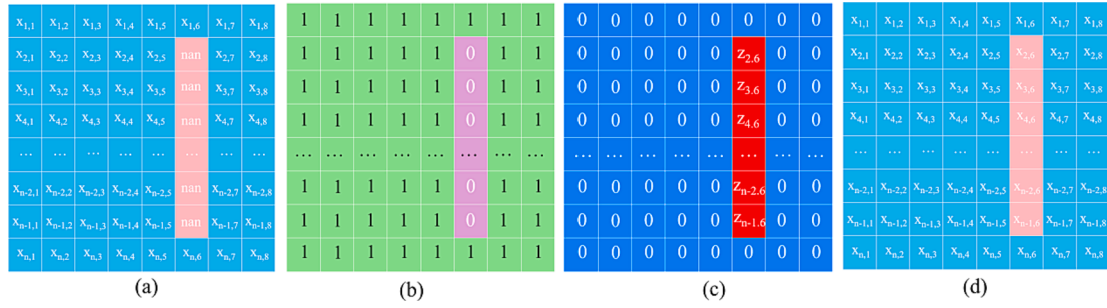


Fig. 2. The input of generator: (a) Incomplete Dataset $\hat{\mathbf{X}}$, (b) Mask $\mathbf{M}$, (c) Random $\mathbf{Z}$; and output of generator (d) Imputed dataset $\bar{\mathbf{X}}$.

This paper proposes a novel data-driven missing data imputation method based on unsupervised GAN frameworks. Because generator and discriminator of the proposed network architecture are slim, which improves the data loss imputation accuracy and efficiency, the slim GAN imputation framework is also called SGAIN. A comparative analysis is carried out with and without consideration of structural spatiotemporal information on a highway-railway dual-purpose bridge in China. This method is also employed to investigate the short-term and long-term continuous structural response under multichannel spatiotemporal information. When SGAIN is compared with the existing imputation method under some valuable scenarios, SGAIN shows a stunning imputation performance and extraordinary efficiency.

2. SGAIN framework for missing data imputation

2.1. Configurations of the proposed SGAIN

In this paper, an imputation method based on unsupervised generative adversarial networks is proposed to solve data loss problems by absorbing the advantages of both generative and discriminative models, which is derived from GAIN [37,38]. It is noted from Fig. 1 that the framework structure of SGAIN mainly includes a generator ($\mathbf{G}$) and discriminator ($\mathbf{D}$). Compared with the state-of-the-art imputation methods, the generator and discriminator are composed of two fully connected layers (FC layers) in SGAIN architecture, which have slimmer structures. In general, the generator generates multiple data probability distributions, the Hint mechanism restricts the generator to generate only one kind of data probability distribution. However, the probability distributions of bridge's dynamic deflection are not complex, and the function of hint mechanism in this application becomes not important. While hint mechanism increases the number of layers of the missing data imputation framework, which increases the computational time. Also, discriminator is needed to identify whether the generated Hint matrix is close to 1 or not, which increases the computational time too. In this paper, hint mechanism is therefore omitted. In the proposed SGAIN framework, the process of missing data imputation is required to extract the features of every data, and finally synthesize the feature information through the fully connected layer. The application of fully connected layers can improve the nonlinear expression ability and learning ability of the model. The SGAIN framework have faster convergence on account of larger derivatives of the Tanh activation function, whereas GAIN utilize sigmoid activation function. The input component of SGAIN can be single-channel data without spatiotemporal information or multiple channel data with spatiotemporal information respectively, consisting of the incomplete data matrix, random matrix, and mask matrix. With the lost data imputation, the generator is trained to generate an imputed data matrix precisely, and discriminator aims to distinguish the recovering data matrix or real data promptly. In detail, mask matrix produces the missing probability of the incomplete data, and the random matrix is represented as the initialization of the lost data. In other words, the generator is responsible for generating missing data, and the

discriminator is responsible for identifying whether the generated data is true or fake. The implemented process is similar to conducting a game theory scenario in which generator and discriminator confront each other.

The purpose of the SGAIN is to estimate every data for incomplete dataset, including missing part and non-missing part. After generator generates the imputed values (both missing and non-missing), the discriminator identifies the imputed values as real or fake. The function of the discriminator is to compare whether the imputed value of the non-missing part matches the original measured value (real or fake). While the measured value of missing part is set to be 0 or NAN, the discriminator has no way to judge them as real or fake because of no reference value. If the discriminator determines fake, the generator will be prompted to continue generating imputed values, and imputation will be optimized with back propagation until the discriminator discriminates real. Finally, the output of SGAIN framework is composed of imputed values of the missing part and measured data of non-missing part.

2.2. Generator

Generator plays a pivotal role in SGAIN framework to generate the data matrix with recovery data through inputting the distribution of multiplied incomplete datasets. The raw incomplete dataset produces an incomplete data matrix, noise matrix, and mask matrix as input.

$$\mathbf{G} : \hat{\mathbf{X}} \times \mathbf{Z} \times \mathbf{M} \times \mathbf{R}^n \rightarrow \bar{\mathbf{X}} \quad (1)$$

Where $\bar{\mathbf{X}}$ denotes the imputed data matrix, $\hat{\mathbf{X}}$ is the incomplete matrix, $\mathbf{Z}$ represents noise matrix, and $\mathbf{M}$ represents mask matrix.

As shown in Fig. 2, the inputs of generator are respectively incomplete dataset $\hat{\mathbf{X}}$, Mask matrix $\mathbf{M}$ and random matrix $\mathbf{Z}$; and output of generator. As shown in Fig. 2 (a), an incomplete data matrix consists of measured data and missing value NAN. The mask matrix identifies the layout of the measured data and the missing data, where the value of measured data and missing data are represented as 1 and 0 respectively. The value at the measured data position is filled with 0, and the value at the missing position is random in the random matrix $\mathbf{Z}$. It is noted from Fig. 2 (d) that missing part and non-missing part of incomplete dataset are estimated by the generator which generates the imputed dataset $\bar{\mathbf{X}}$. As demonstrated in Eqs. (2) and (3), the generating process of generator is closer to the one used in GAIN [37].

$$\mathbf{Z} = \mathbf{M} \odot \hat{\mathbf{X}} + (1 - \mathbf{M}) \odot \mathbf{N} \quad (2)$$

$$\bar{\mathbf{X}} = \mathbf{G}(\mathbf{Z}, \mathbf{M}) \quad (3)$$

Where $\bar{\mathbf{X}}$ denotes the imputed data matrix, $\hat{\mathbf{X}}$ is the incomplete matrix, $\mathbf{Z}$ represents random matrix, $\mathbf{M}$ represents mask matrix, $\mathbf{N}$ is random value of uniform distribution and $\odot$ represents element-wise multiplication.

In the equation, $1 - \mathbf{M}$ makes the original value being 0, ensuring that

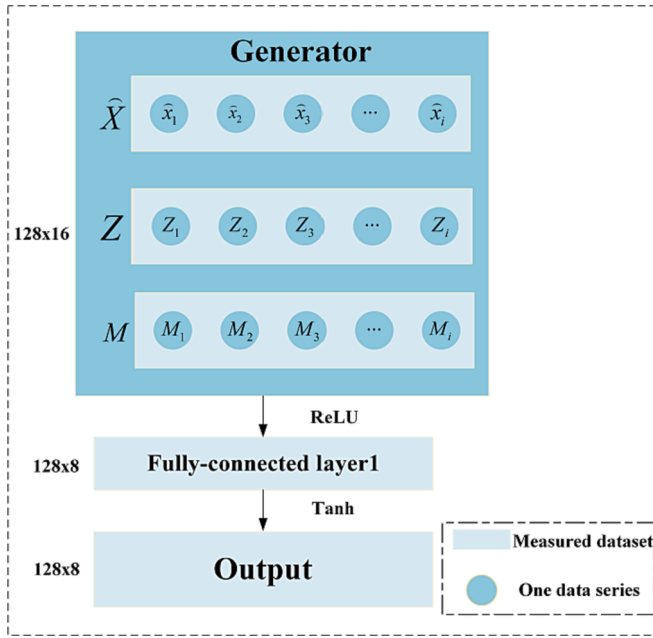


Fig. 3. The generator adopted in this paper.

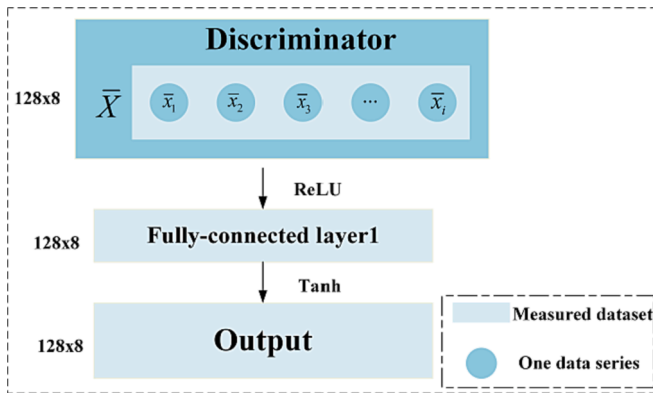


Fig. 4. The discriminator structure adopted in this paper.

noise values are located in the missing data, and generator reflects noise values into the data space. $\bar{X}$ is imputed data, which ensure that the imputation data can be similar to the measured value. In addition, the features of the non-missing data can be extracted to improve the accuracy and efficiency of the proposed method.

$$\hat{X} = M \odot \bar{X} + (1 - M) \odot \bar{X} \quad (4)$$

Where $\hat{X}$ denotes the data generated by the generator, composing the value of the measured part $M \odot \bar{X}$ and the imputed value of the missing part. That being said, even if the original measured data is generated by the SGAIN algorithm, the non-missing component in data remains lastly, which is the function of the generator is in the same way as in the GAIN framework. As shown in Fig. 3, it is noted that generator structure is composed of the two full-connection layers, which have a slim structure to improve the executing efficient of proposed method. The ReLU and Tanh activation functions embedded in full-connection layers create a stable implement.

The input of the fully connected layer is a multidimensional array, which is flattened for one-dimensional processing, and is connected to the fully connected layer. To combine the features extracted from the previous layer, each node of the fully connected layer is connected to every node of the previous layer. The structure of the fully connected layer is shown in Fig. 3. In addition, the fully connected layer of the generator is described in the equation (5).

$$\bar{x}_i = f \left(W_{i1} * \hat{x}_1 + W_{i2} * \hat{x}_2 + W_{i3} * \hat{x}_3 + \dots + W_{ii} * \hat{x}_i \right) \quad (5)$$

where $\hat{x}$ is the input of Fully Connected Layer; $\bar{x}_i$ is output of Fully Connected Layer; W_{ii} is the weight coefficient; f is the activation functions, including ReLU and Tanh.

2.3. Discriminator

The purpose of the discriminator is to distinguish imputation data from real data. Similarly, to achieve training conciseness and efficiency, fully connected structures are utilized in the discriminator, the configuration of the discriminator is demonstrated in Fig. 4. With guaranteed calculation accuracy, the compact structure improves training efficiency and accelerates the training process. The stability of the SGAIN model can be improved if the mask matrix substitutes the softening value (e. g. 0.9 or 0.1) for the label value of 1 or 0. In addition, a random noise matrix enhances the stability and computation speed, also alleviating the mode descent and mode collapse. The input of the discriminator is produced by the generator, namely $\bar{X}$ data matrix, in Fig. 2 (d). As shown

$r_{1,1}$	$r_{1,2}$	$r_{1,3}$	$r_{1,4}$	$r_{1,5}$	$r_{1,6}$	$r_{1,7}$	$r_{1,8}$
$r_{2,1}$	$r_{2,2}$	$r_{2,3}$	$r_{2,4}$	$r_{2,5}$	$r_{2,6}$	$r_{2,7}$	$r_{2,8}$
$r_{3,1}$	$r_{3,2}$	$r_{3,3}$	$r_{3,4}$	$r_{3,5}$	$r_{3,6}$	$r_{3,7}$	$r_{3,8}$
$r_{4,1}$	$r_{4,2}$	$r_{4,3}$	$r_{4,4}$	$r_{4,5}$	$r_{4,6}$	$r_{4,7}$	$r_{4,8}$
...	...	...	...	...	...	...	...
$r_{n-2,1}$	$r_{n-2,2}$	$r_{n-2,3}$	$r_{n-2,4}$	$r_{n-2,5}$	$r_{n-2,6}$	$r_{n-2,7}$	$r_{n-2,8}$
$r_{n-1,1}$	$r_{n-1,2}$	$r_{n-1,3}$	$r_{n-1,4}$	$r_{n-1,5}$	$r_{n-1,6}$	$r_{n-1,7}$	$r_{n-1,8}$
$r_{n,1}$	$r_{n,2}$	$r_{n,3}$	$r_{n,4}$	$r_{n,5}$	$r_{n,6}$	$r_{n,7}$	$r_{n,8}$

(a)

$f_{1,1}$	$f_{1,2}$	$f_{1,3}$	$f_{1,4}$	$f_{1,5}$	$f_{1,6}$	$f_{1,7}$	$f_{1,8}$
$f_{2,1}$	$f_{2,2}$	$f_{2,3}$	$f_{2,4}$	$f_{2,5}$	$f_{2,6}$	$f_{2,7}$	$f_{2,8}$
$f_{3,1}$	$f_{3,2}$	$f_{3,3}$	$f_{3,4}$	$f_{3,5}$	$f_{3,6}$	$f_{3,7}$	$f_{3,8}$
$f_{4,1}$	$f_{4,2}$	$f_{4,3}$	$f_{4,4}$	$f_{4,5}$	$f_{4,6}$	$f_{4,7}$	$f_{4,8}$
...	...	...	...	...	...	...	...
$f_{n-2,1}$	$f_{n-2,2}$	$f_{n-2,3}$	$f_{n-2,4}$	$f_{n-2,5}$	$f_{n-2,6}$	$f_{n-2,7}$	$f_{n-2,8}$
$f_{n-1,1}$	$f_{n-1,2}$	$f_{n-1,3}$	$f_{n-1,4}$	$f_{n-1,5}$	$f_{n-1,6}$	$f_{n-1,7}$	$f_{n-1,8}$
$f_{n,1}$	$f_{n,2}$	$f_{n,3}$	$f_{n,4}$	$f_{n,5}$	$f_{n,6}$	$f_{n,7}$	$f_{n,8}$

(b)

Fig. 5. The output of discriminator: (a) Real matrix, (b) Fake matrix.

in Fig. 5, the output of the discriminator is either a real or fake data matrix. For more robustness, the optimizer of the SGAIN framework adopts the Stochastic Gradient Descent (SGD).

Meanwhile, the fully connected layer of the discriminator is described in the equation (6).

$$\hat{x}_i = f(W_{i1} * \bar{x}_1 + W_{i2} * \bar{x}_2 + W_{i3} * \bar{x}_3 + \dots W_{ii} * \bar{x}_i) \quad (6)$$

where $\bar{x}_i$ is the input of Fully Connected Layer; $\hat{x}_i$ is output of Fully Connected Layer; W_{ii} is the weight coefficient; f is the activation functions, including ReLU and Tanh.

2.4. Objective and evaluation criteria

In this study, the SGAIN framework has two important purposes: the purpose of discriminator D is to maximize the probability of correctly predicting the $\mathbf{M}$ matrix; the purpose of the generator is to minimize the probability of D's predicted $\mathbf{M}$ matrix. Moreover, the back propagation is utilized in SGAIN to optimize the generator and discriminator. In detail, the loss is between non-missing part of measured data ($\hat{\mathbf{X}}$) and non-missing part of imputed data ($\hat{\mathbf{X}}$) generated by generator. To improve the imputation efficiency and accuracy, the discriminator and generator are optimized by the objective function (7) and (8).

$$\max_D L(\alpha_D) = E_{\hat{\mathbf{X}}, \mathbf{M}} [\mathbf{M} \odot D(\hat{\mathbf{X}}) + (1 - \mathbf{M}) \odot (1 - D(\hat{\mathbf{X}}))] \quad (7)$$

$$\min_G L(\alpha_G) = -E_{\hat{\mathbf{X}}, \mathbf{M}} [(1 - \mathbf{M}) \odot D(\hat{\mathbf{X}})] + \alpha E \left[(\mathbf{M} \odot \hat{\mathbf{X}} - \mathbf{M} \odot \bar{\mathbf{X}})^2 \right] \quad (8)$$

where $\hat{\mathbf{X}}$ denotes the data generated by the generator, composing the value of the non-missing part and the imputed value of the missing part; $\bar{\mathbf{X}}$ denotes the imputed data matrix; $\hat{\mathbf{X}}$ is the incomplete matrix; $\mathbf{M}$ represents mask matrix; and α represents hyperparameters.

To evaluate the performance of proposed model comprehensively, the correlation coefficients (R^2), root mean square error (RMSE), and mean absolute error (MAE) are presented as available quantitative performance metrics in the time domain. In addition, accuracy and the mean negative log-likelihood (MNLL) are separately introduced to assess the predicted accuracy and robustness of the model. Meanwhile, the good model should have a high R^2 value, a low RSME value, a low MAE value, a high accuracy index, and a high MNLL. These performance metrics can be expressed as:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y'_i - y_i)^2}{\sum_{i=1}^n (\bar{y} - y_i)^2} \quad (9)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |y'_i - y_i| \quad (10)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - y_i')^2} \quad (11)$$

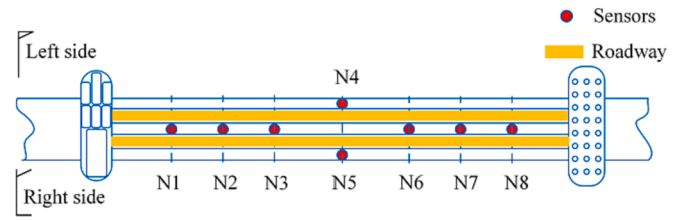
$$MNLL = -\log(P(y_i | y_i, \sigma_{y_i}^2)) = \frac{1}{n} \sum_{i=1}^n \left(\frac{1}{2} \log(2\pi \sigma_{y_i}^2) + \frac{(y_i - y_i')^2}{2\sigma_{y_i}^2} \right) \quad (12)$$

$$Accuracy = 1 - \frac{\|y_i - y'_i\|_2}{\|y'_i\|_2} = 1 - \frac{\sqrt{\sum_{i=1}^n (y_i - y'_i)^2}}{\sqrt{\sum_{i=1}^n y_i'^2}}$$

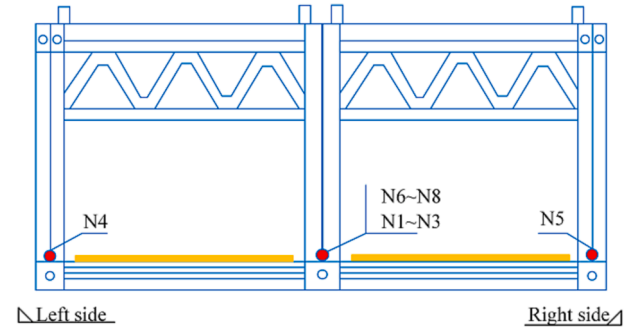
Where y_i denotes the measure values of i th location, y'_i represents the imputation values of i th location, $\sigma_{y_i}^2$ is the variance of the imputation values, $\bar{y}$ demonstrates the mean of the imputation values.



(a)



(b)



(c)

Fig. 6. The sensors layout of the bridge.

3. Implementation and analysis

To assess the imputation capability of SGAIN framework, the practical applications are conducted on the deflection of a highway-railway dual-purpose bridge. The imputation of 8 crucial fault sensors with several different missing scenarios is implemented with SGAIN model. The imputation framework is applied in TensorFlow, and the configurations of the computational platform are two Intel Xeon(R) E5-2696 v4 CPUs, a 256 GB memory, and an NVIDIA TITAN X (Pascal) GPU for boosting algorithm application. The implemented process of the data imputation method will be discussed in detail below.

3.1. Dataset description

The proposed imputation method is validated by comparing the imputed deflections with the measured ones of Tongling Yangtze River Bridge, which is a long-span highway-railway dual-purpose bridge in China. As shown in Fig. 6(a), the main bridge is a typical five-span steel truss cable-stayed bridge which carried out important road and railways transport. The deflection of the main span caused by vehicle load and ambient load was measured at the sampling frequency of 5 Hz. There are

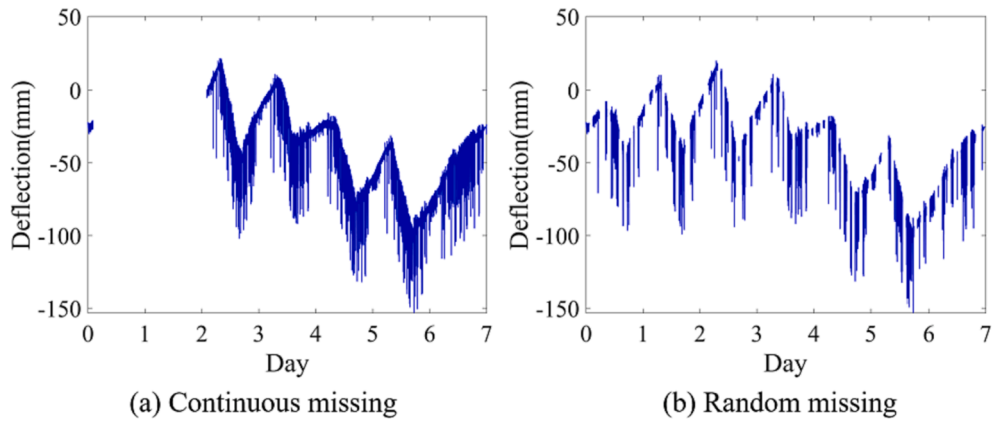


Fig. 7. Data loss patterns (a) continuous data loss (b) random data loss.

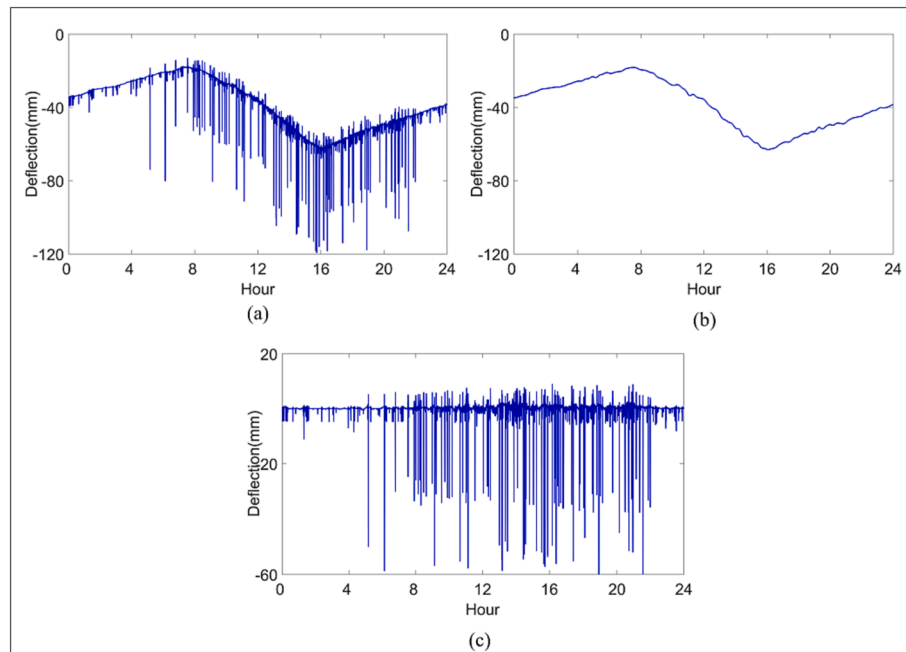


Fig. 8. Extraction of different excitation-induced deflection from raw measurements: (a) raw measurement of N6, (b) temperature-induced deflection, and (c) dynamic-induced deflection.

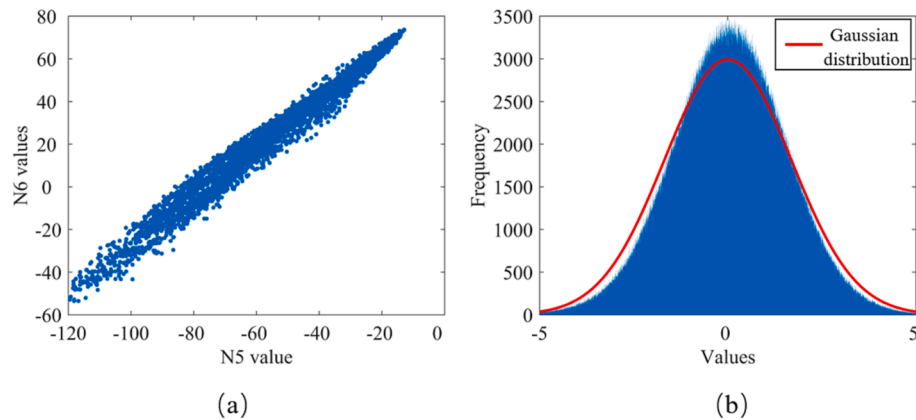
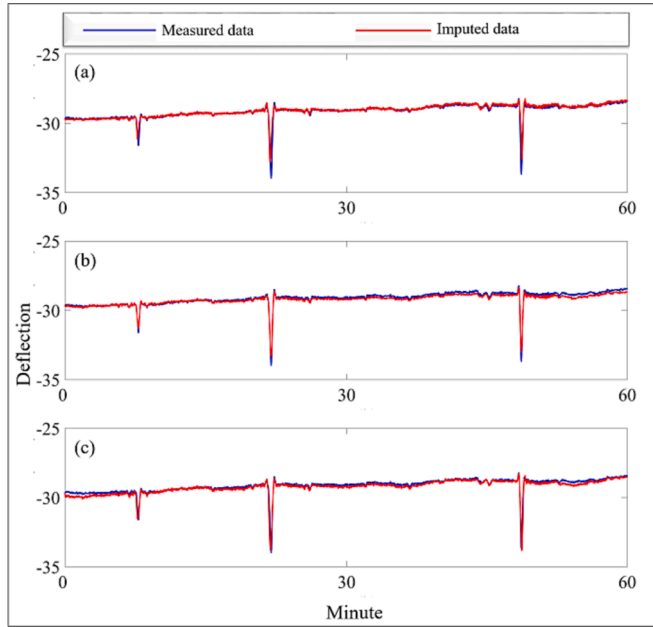


Fig. 9. (a) The linear relationship between sensors (b) Distribution of target sensor (N6).

Table 1

The PCCs between N6 sensor and other sensors.

Criteria	N1	N2	N3	N4	N5	N7	N8
PCC	0.6879	0.8519	0.9323	0.9827	0.9804	0.9743	0.9193
R ²	0.4732	0.7258	0.8692	0.9658	0.9612	0.9493	0.845

**Fig. 10.** The influence of input dataset correlation on missing data imputation (a) Level-I (b) Level-II, (c) Level-III.

a total of 8 deflection sensors (numbered N1–N8) in main span of the bridge which are adopted for practical applications of SGAIN framework, as shown in Fig. 6(b) and (c). In general, missing data can be categorized into random missing (RM) and continuous missing (CM) [14], and the data loss patterns of Tongling Yangtze River Bridge are illustrated in Fig. 7 (a) and (b) respectively. There are mainly three missing data imputation scenarios in this paper, namely single-sensor random loss without considering spatiotemporal correlation (one-to-one) as type I, single-sensor random loss considering spatiotemporal correlation (many-to-one) as type II, and single-sensor continuous loss considering spatiotemporal correlation (many-to-one) as type III. The one-to-one model could be defined when input and output of the network only have one single sensor. Many-to-one means that the input contains multiple sensors (other sensors) while the output has only a single sensor which is for the fault sensor (N6) in this paper. For validating the imputation performance of SGAIN, the sensor (N6) is selected as the main target fault data sensor with several different loss ratios (ranging from 1 day to 7 days), and other deflection sensors are exploited as the training dataset.

3.2. Effect of the dataset correlation

The correlation between sensors plays a significant role in investigating the imputation performance of SGAIN. To capture the underlying

Table 2

The imputation accuracy of three-level.

Types	Sensors	1 h	2 h	3 h	4 h	5 h	6 h	7 h
Level-I	N4,N5,N7	0.9977	0.986	0.9834	0.9827	0.9758	0.9735	0.9797
Level-II	N1,N2	0.9978	0.9886	0.9853	0.9807	0.9738	0.9642	0.9663
Level-III	N1–N8	0.9854	0.9828	0.9831	0.9744	0.9737	0.9623	0.9406

relationships between bridge's deflection sensors accurately, the raw deflections are decomposed into deflection induced by ambient temperature and deflection induced by the dynamic excitation using the wavelet decomposition. Respectively, the measured deflection induced by different excitation is shown in Fig. 8.

The Pearson correlation coefficient can be used to measure the degree of linear correlation between the data if: (1) The two data are linearly related to each other and both are continuous; (2) The data conform to Gaussian distribution; (3) The two data are independent of each other. It is well known from the central limit theorem that when the number of data is sufficiently numerous, the limit of the data distribution is the Gaussian distribution (normal distribution). In this paper, the bridge deflection was imputed by considering the signal from sensors at the same girder. As shown in Fig. 9, it can be found that the deflection data on this bridge satisfy the linear relationship and the Gaussian distribution essentially, the Pearson coefficient is therefore applied in this paper. In some other non-linear and non-Gaussian distributed data case which may happens in some complicated conditions, the Kendall correlation coefficient or Spearman correlation coefficient will then be considered.

The Pearson correlation coefficient (PCC) is used to determine the relationship between target sensor (N6) and other sensors. Table 1 shows the PCCs between target sensor N6 and other sensors. Based on PCC results, in this paper, the input sensors of SGAIN are set as three types to study the missing data imputation: the high-grade correlation of sensors (N4, N5, and N7) is designated as Level-I ($PCC > 0.95$); the middle correlation of sensors (N1 and N2) is defined as Level-II ($0.6 < PCC < 0.9$); the fusion correlation of sensor (N1–N8) means Level-III ($0 < PCC < 1$). In addition, data augmentation is introduced to keep the consistency of the numbers of sensors in the training SGAIN model process.

In order to assess the imputation performance under the three-level input scenarios, N6 data imputation with different continuous missing rates (ranging from 1 h to 7 h) is implemented on the SGAIN framework, considering spatiotemporal correlation (Type III). It can be noted from Fig. 10 that the SGAIN model has outstanding imputation results under three-level scenarios, and the imputation curves trend are similar in 1 h data loss essentially. Table 2 shows these imputation results obtained by proposed imputation method. Interestingly, the performance of Level-I and Level-II slightly outperforms the Level-III dataset from 1 h to 7 h data loss. Since the SGAIN framework extracts limited authentic information in the mixture features of Level-III, the imputation result of Level-III is less accurate than Level-I and Level-II. In the long-term monitoring of the bridge, the mixed dataset with various correlations between sensors is common. Thus, based on the most unfavorable principle, Level-III is chosen to conduct the data imputation, which is considered the spatiotemporal (many-to-one), on SGAIN in this paper.

3.3. Single-sensor imputation (one-to-one)

The random missing data are usually resulted from sensor

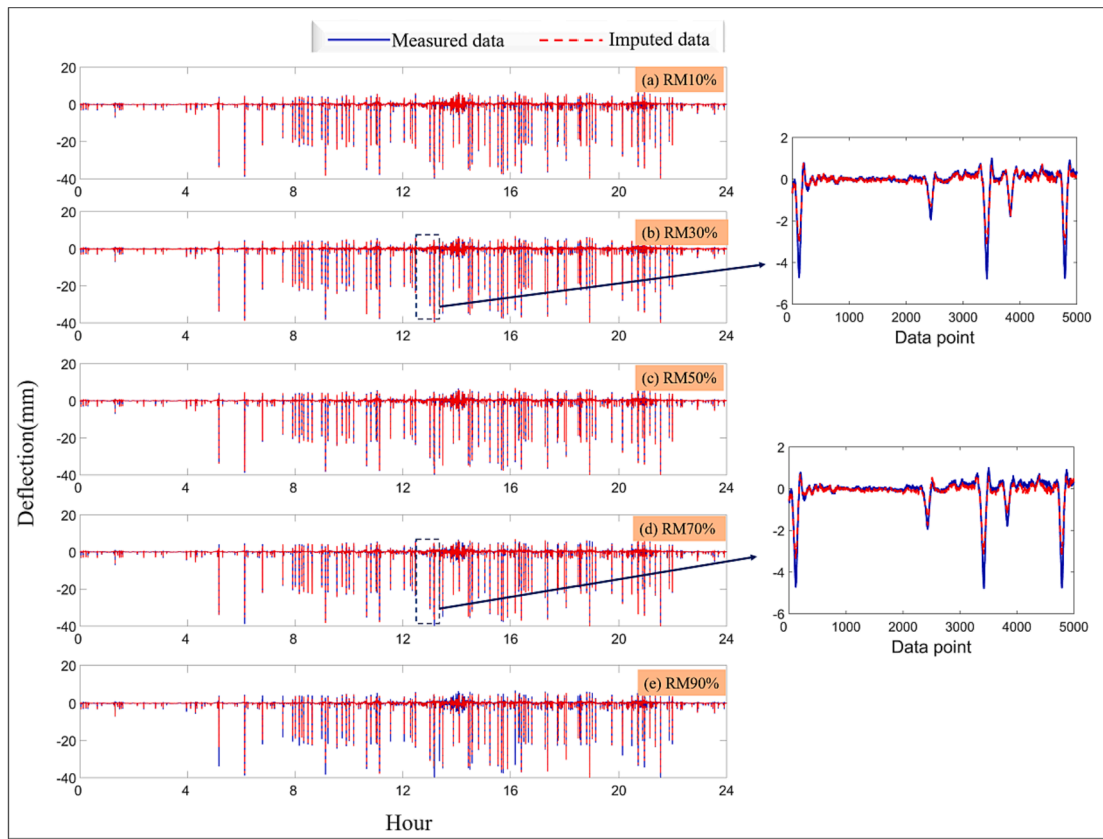


Fig. 11. Imputation performance of one-to-one under different loss ratios (Type I).

Table 3
Accuracy for the SGAIN-based approach (one to one).

Sensor	10 %	30 %	50 %	70 %	90 %
N1	0.985	0.961	0.947	0.9276	0.9036
N2	0.9875	0.9724	0.9376	0.9444	0.9161
N3	0.984	0.93	0.906	0.8661	0.849
N4	0.9964	0.9658	0.942	0.926	0.8802
N5	0.995	0.9820	0.9589	0.9223	0.8917
N6	0.9968	0.9761	0.9678	0.9363	0.9081
N7	0.978	0.9656	0.935	0.905	0.862
N8	0.9353	0.9241	0.9027	0.8933	0.8412

disconnection or poor contact occasionally in the SHM systems. Single-sensor random loss without considering spatiotemporal correlation (one-to-one) is defined as type I. This paper firstly studies the imputation results of 1-day random missing data with different missing ratios

ranging from 10 % to 90 %. Fig. 11 shows the imputation results with the measured deflection data of the bridge SHM by the SGAIN model. As can be seen from Fig. 11(a) to (h), SGAIN effectively recovers the bridge deflection data by dynamic excitation. Circumstantially, the imputation data almost overlaps the measured data only with a very small deviation, which is within the controllable range. Moreover, as Table 3 shows, it is apparent that the imputation accuracy for the SGAIN-based approach is remarkable in 8 fault sensors respectively. This illustrates that SGAIN can extract the amounts of information from non-missing parts of fault sensors, even in severe scenarios. In addition, when the missing area increases, the data imputation ability slightly degenerates as the accuracy emerging descending trend.

Compared with the traditional linear interpolation method, SGAIN estimates every data in the dataset, and continuously optimizes the generated values, finally outputs the complete data. SGAIN learns the data distributed feature based on the remaining measured data, for the

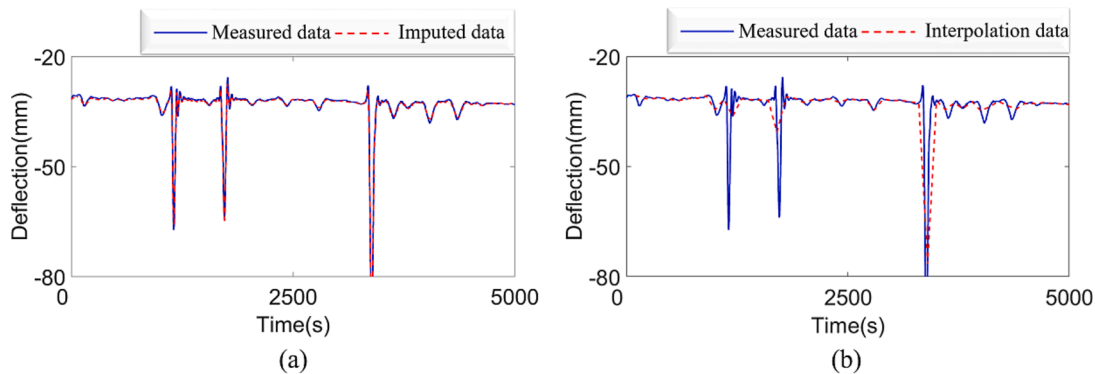


Fig. 12. the comparison between SGAIN and simple linear interpolation. (a) SGAIN; (b) linear interpolation method.

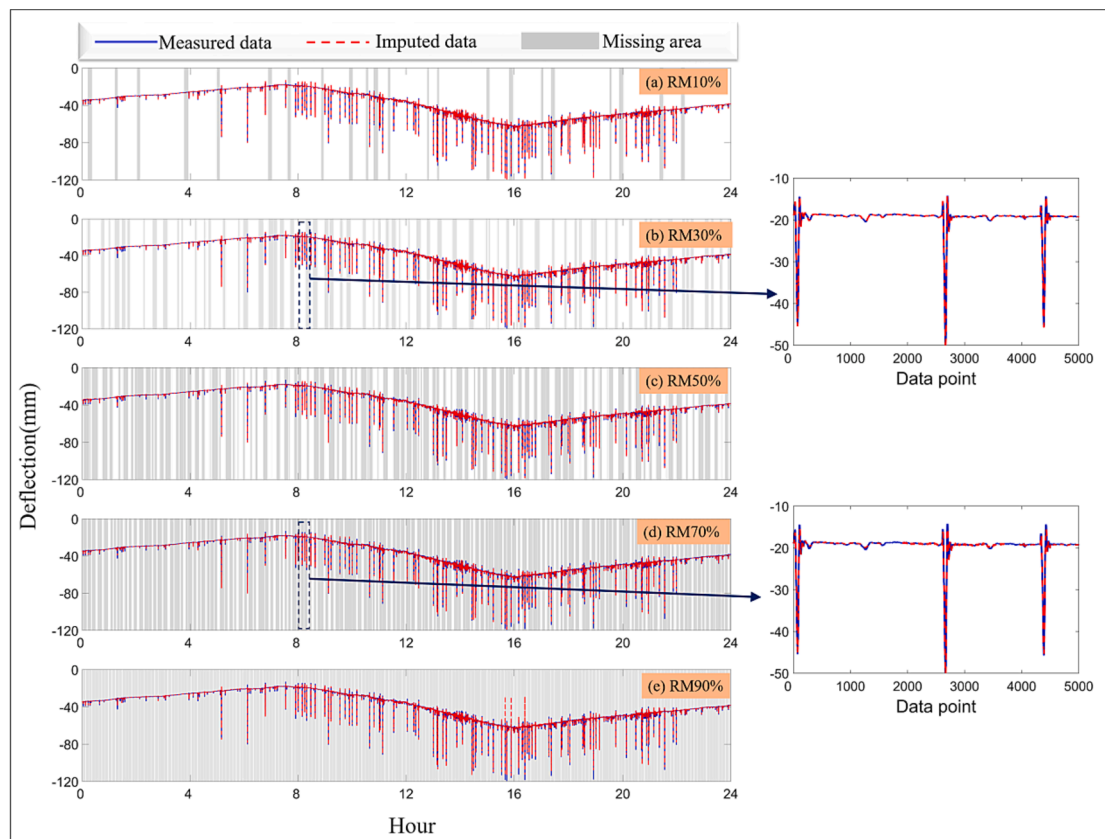


Fig. 13. 1-day response imputation of sensor N6 with Type-II scenario.

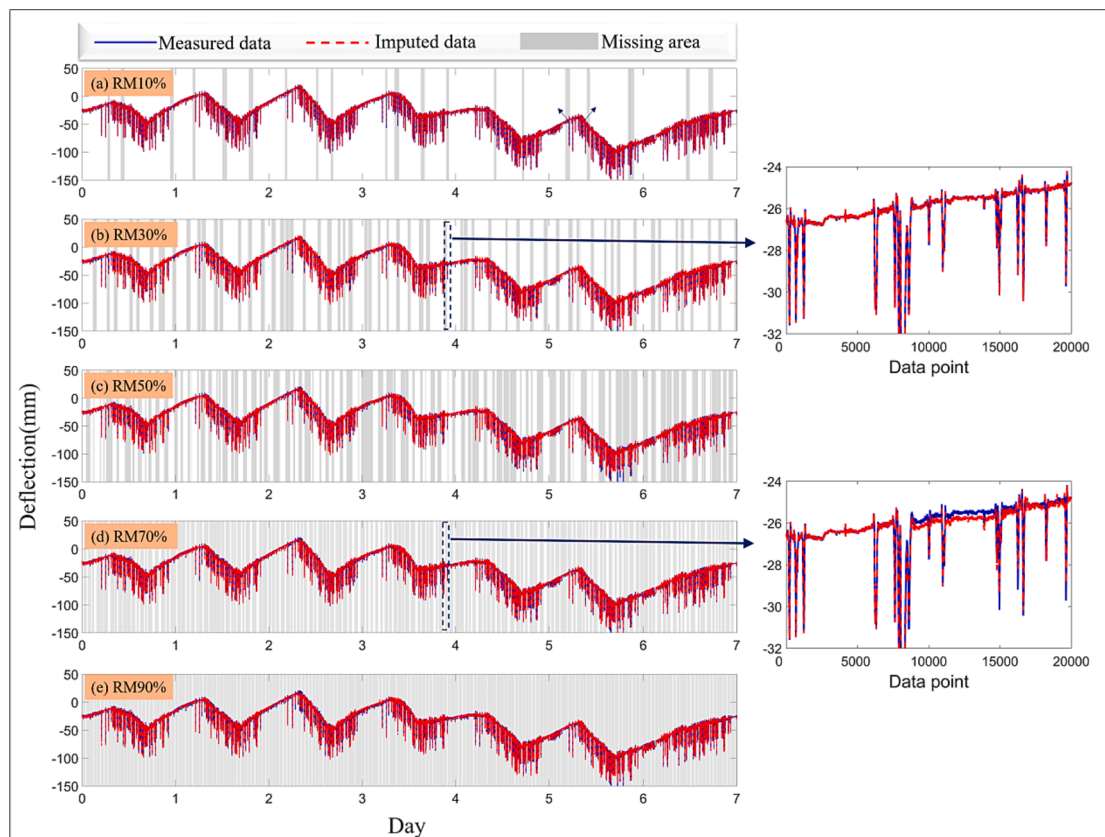


Fig. 14. 7-day response imputation of sensor N6 with Type-II scenario.

Table 4
The performance metrics of 10–90% random missing ratio in 7-day.

Criteria	Missing ratio				
	10 %	30 %	50 %	70 %	90 %
MAE	0.5843	0.6264	0.6922	0.6929	0.7175
RMSE	0.7788	0.8039	0.8589	0.8795	0.898
R2	0.9991	0.999	0.9989	0.9987	0.9981
MNLL	4.1656	4.1817	4.1987	4.2029	4.2297
Error	0.0187	0.0208	0.0381	0.0456	0.0566
Accuracy	0.9813	0.9792	0.9619	0.9544	0.9434

missing data generation and discrimination. However, the linear interpolation method uses a straight line connecting two known quantities to determine the value of an unknown quantity between the two known quantities. Such method is relatively simple and can't completely extract the data information to improve the interpolated accuracy. As shown in Fig. 12, the imputed performance of SGAIN framework is clearly superior to the simple linear interpolation method.

3.4. Single-channel random imputation (many-to-one)

In SHM systems, various correlated deflection sensors are embedded to monitor load-bearing capacity of bridges. The spatiotemporal correlation among sensors can significantly contribute to imputation performance of random missing data ranging from 1 day to 7 days, namely missing type II (many-to-one). In previous 3.2 section, imputation effect of the sensors' correlation is studied. Based on the most unfavorable principle, Level-III is chosen to conduct the data imputation on the scenario considering the spatiotemporal (many-to-one). That is, the

channel 6 (N6) is set to be the fault sensor, and the channel 1, 2, 3, 4, 5, 7, 8 in main span are set as SGAIN inputs. As can be seen from Fig. 13, the imputation effect of the proposed framework is primarily studied on the 1-day random missing deflection data with different missing ratios (e.g., 10 % to 90 %). The comparison results between imputation and real are shown in gray area. What stands out in Fig. 13 is that the SGAIN maintain good imputation performance, especially with high data loss ratios (e.g., 90 %). Whereas, as shown in Fig. 13 (e), the slight error on 16 h is owing to a high missing proportion and the vanishing gradients.

The imputation effect of random missing on 1-day manifests remarkable performance. However, the bridge SHM system always collects and stores the monitoring data. Since the sample rate of deflection data is set as 5 Hz and the amount of data in a week is 7 times of that in 1-day, the training data volume and the high loss ratio in 7-day data will be a challenge to the implemented accuracy and efficiency of SGAIN model. In this paper, long-term imputation with 7-days random missing is studied to evaluate the robustness and generalization capacity of SGAIN imputation. Fig. 14 shows the imputation results of the SGAIN model with the missing ratios being 10 %, 30 %, 50 %, 70 %, and 90 % respectively. Even if the missing ratio reaches up to 90 %, the imputation effect of the SGAIN framework provides outstanding results. The reason is that the proposed method can extract the non-missing part of fault sensor, and simultaneously absorb spatiotemporal correlations from other sensors, increasing the learning competence of SGAIN imputation.

For quantitative analysis, six performance criteria, including MAE, RMSE, R^2 , MNLL, Error and Accuracy, are introduced to evaluate the imputation ability of proposed imputation method. In contrast, R^2 and Accuracy decrease continuously when the missing ratio increases in 7-day, which is shown in Table 4. Furthermore, as the missing rate

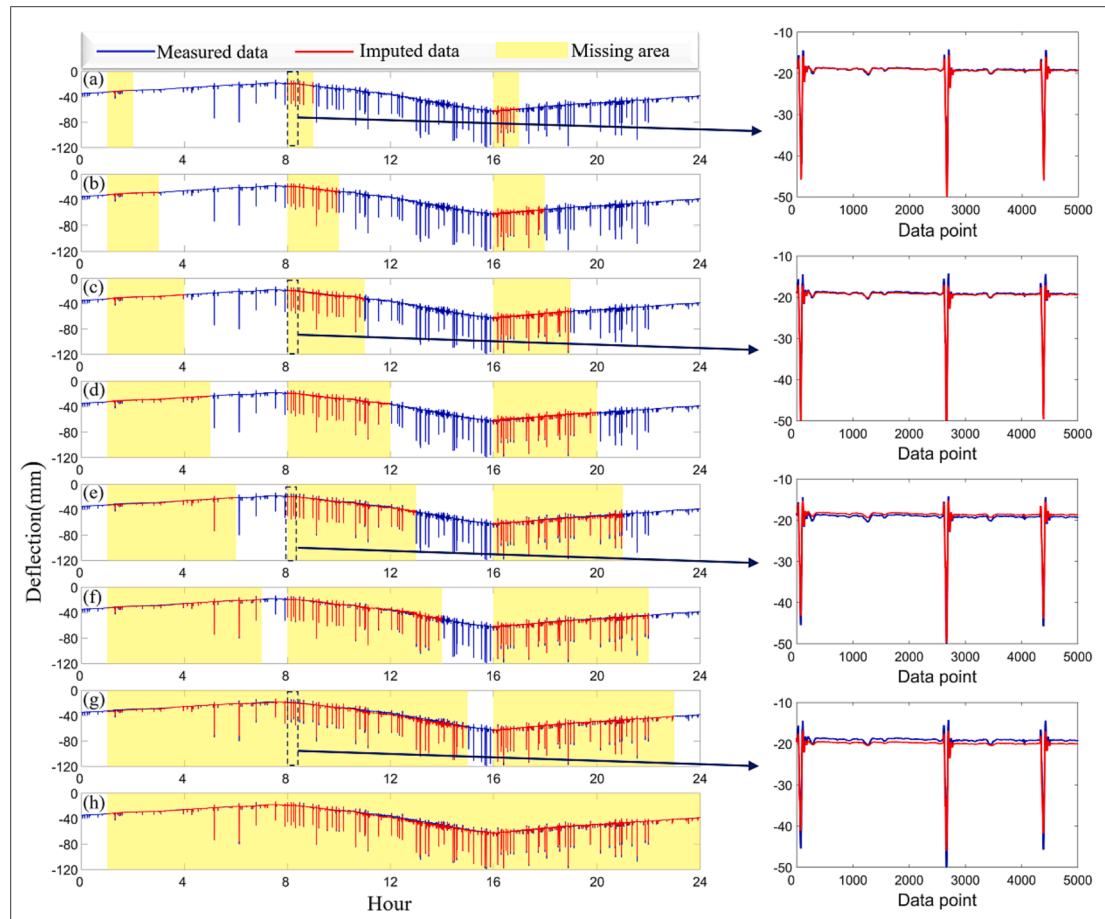


Fig. 15. 1-day Response imputation of channel N6 within Type-III scenario.

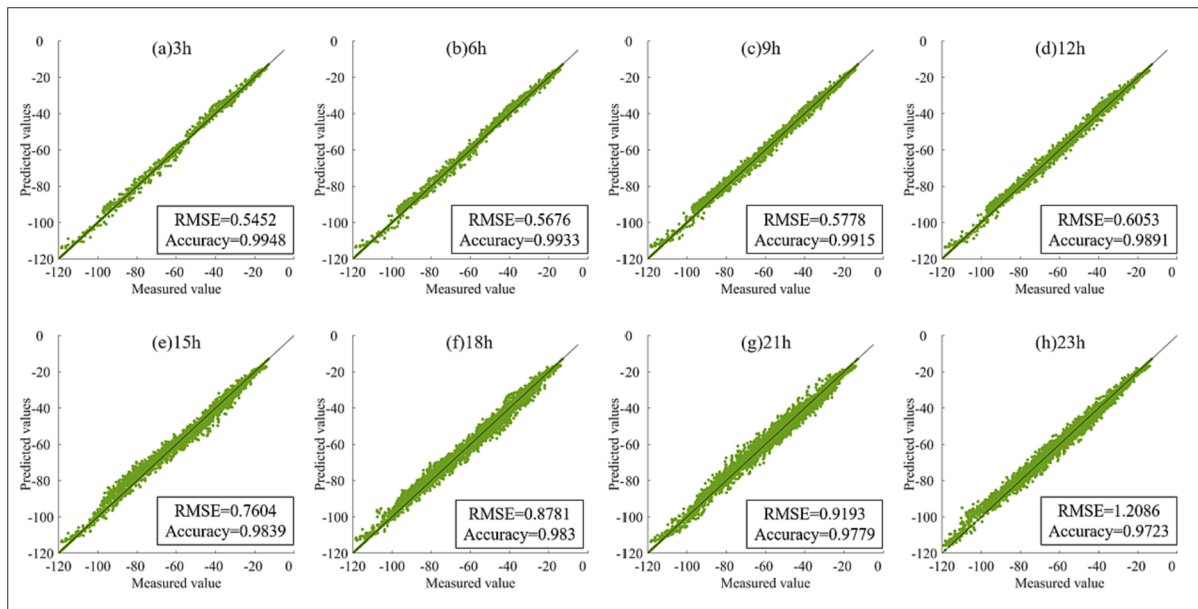


Fig. 16. Correlations between the measured and predicted values under 3–23 h continuous missing scenario.

increase continuously, the error of the model accumulates significantly. To further analyze the mode of error accumulation, the incremental ratios of MAE, RMSE, and R^2 of the model at the missing ratios between 10 % and 30 %, 30–50 %, 50–70 % and 70–90 % are calculated. The incremental ratios of MAE are 7.21 %, 10.5 %, 1.01 % and 3.55 %, while RMSE increments are 3.22 %, 6.84 %, 2.4 % and 2.1 % respectively. The results show the growth ratio at the period between 30 % and 50 % is the largest, indicating that the error accumulation speed is the fastest for random missing case.

3.5. Continuous missing data imputation (many to one)

So far, this paper has focused on random missing data imputation. This section will discuss continuous missing data imputation with SGAIN. In the previous section, the imputation performances of SGAIN model are analyzed on various loss patterns of 1-day and 7-day random missing data. In order to determine the generalization capability of SGAIN framework, it is worthy to investigate whether the imputation method can perform efficiently on the type III scenario that single-sensor continuous loss considering spatiotemporal correlation (many-to-one).

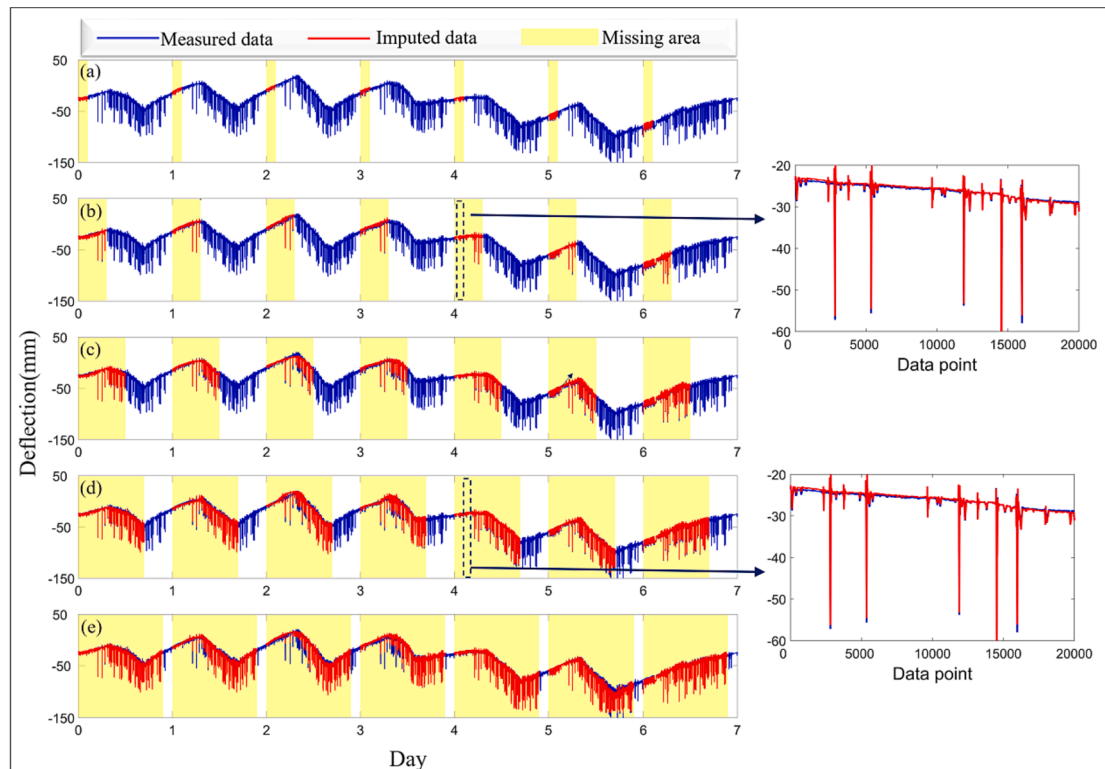


Fig. 17. Response imputation of sensor N6 of an arbitrarily selected 7 days under 10–90 % continuous missing scenario.

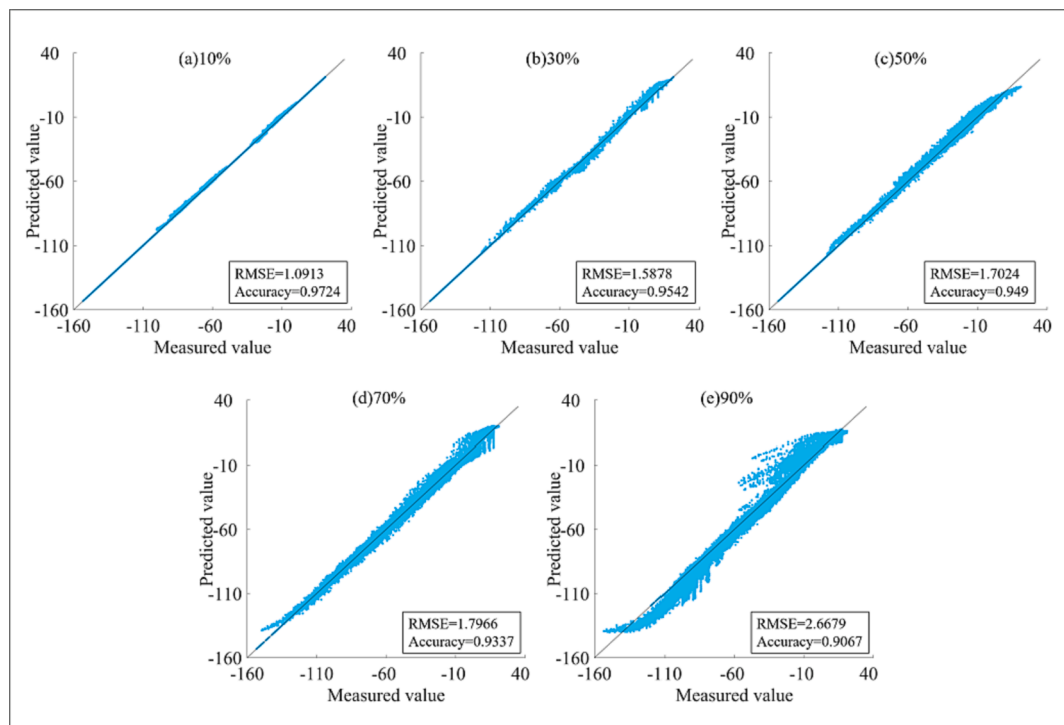


Fig. 18. Correlations between the measured and predicted values of an arbitrarily selected 7 days under 10–90 % continuous missing scenario.

The training data volume and the high loss ratio in 1-day and 7-day data will be a challenge to the implementation of SGAIN model.

Fig. 15 shows the imputation results of measured deflection with continuous missing by SGAIN framework. From Fig. 15(a)–(h), it can be found that 3–23 h of continuous missing data are recovered, in which, the highlighted yellow part and the white part represent the missing and non-missing part respectively. In the yellow part individually, the imputation data closely resembles the original measured data, indicating that the SGAIN method is stable without model collapse and has high accuracy of imputation. Nevertheless, it can be seen from the missing area that the measured line does not cover the imputation line in some aspects. Correspondingly, the discrepancy between the measured and imputed data becomes larger when the missing area gets larger.

Fig. 16 presents the correlation between measured and imputed values for 3–23 h missing. RMSE or Accuracy are applied to fully reflect the prediction ability of the model with various missing rates. The measured and predicted values are highly fitted in linear regression, and the correlation between the measured and predicted values are evenly distributed on the bilateral region of the linear regression curve. In addition, based on the learned latent features, the accurate estimation and forecasting results show the satisfactory performance of the proposed SGAIN approach for processing incomplete SHM recordings, with better quantification capability.

In this paper, we take full effective advantage of SGAIN to compare the data imputation effects of continuous missing, with missing ratios are determined uniformly at 10–90 % for 7 days' deflection data of the bridge. The imputation effect of continuous missing in the 7-day scenario is analyzed separately with 10 %, 30 %, 50 %, 70 %, and 90 % missing ratios. As shown in Fig. 17 10–90% of continuous missing data where highlighted in yellow are recovered well. It demonstrates that with the increase of the missing ratio, the imputation error tends to become larger but with a limited amount. These results also reveal that the SGAIN model has good imputation performance under the condition of a high proportion of data missing.

Fig. 18 elaborates on the correlation and loss between measured data and predicted data. The RMSE or Accuracy are appropriate performance metrics to evaluate the correlations between the measured and predicted

Table 5

The performance metrics of 10–90% continuous missing ratio in 7-day.

Criteria	Missing ratio				
	10 %	30 %	50 %	70 %	90 %
MAE	0.9039	1.2381	1.347	1.3822	1.9591
RMSE	1.0913	1.5878	1.7024	1.7966	2.6679
R2	0.9983	0.9963	0.9952	0.9952	0.9898
MNLL	4.2384	4.3168	4.5179	4.6058	4.7659
Error	0.0276	0.0458	0.051	0.0663	0.0933
Accuracy	0.9724	0.9542	0.949	0.9337	0.9067

values of 7 days under 10–90 % continuous missing scenario. It can be seen that the SGAIN imputation frame has an excellent performance with the missing ratios from 10% to 50 %. The fitting curve of the correlation partially deviates from the positive linear regression curve, especially for the 90 % missing proportion case. As shown in Figs. 17(e) and 18 (e), there is no significant deviation in the imputation effect even when the missing area is very large, indicating the stability of the SGAIN frame in a high missing proportion case. The detailed performance assessment and accuracy measurements are clarified in the comparison below.

As shown in Table 5, the generalization ability of the SGAIN model is evaluated by the performance criteria. In the cases of 10–90 % missing in 7-day, the prediction error and RMSE are both small. It can also be seen that the value of R^2 (closing to 1) in continuous missing imputation is high, indicates that the recovery data fit well with real measured data. Among 7-day continuous missing cases, the increment ratios of MAE are 36.97 %, 8.8 %, 2.6 %, 41.74 %, and RMSE increment ratio are 45.5 %, 7.21 %, 5.53 %, 48.4 % respectively. It is found that the increment between 10 % and 30 % and 70–90 % missing ratios achieved the largest, illustrating that faster errors accumulation in the early and late periods for the continuous missing case. Specifically, the SGAIN model with low missing ratio achieves better imputation performance, which can be verified by the lower MNLL value compared with the other missing ratios. These quantitative results proved that the proposed method is effective in the deflection missing data imputation, and have a strong

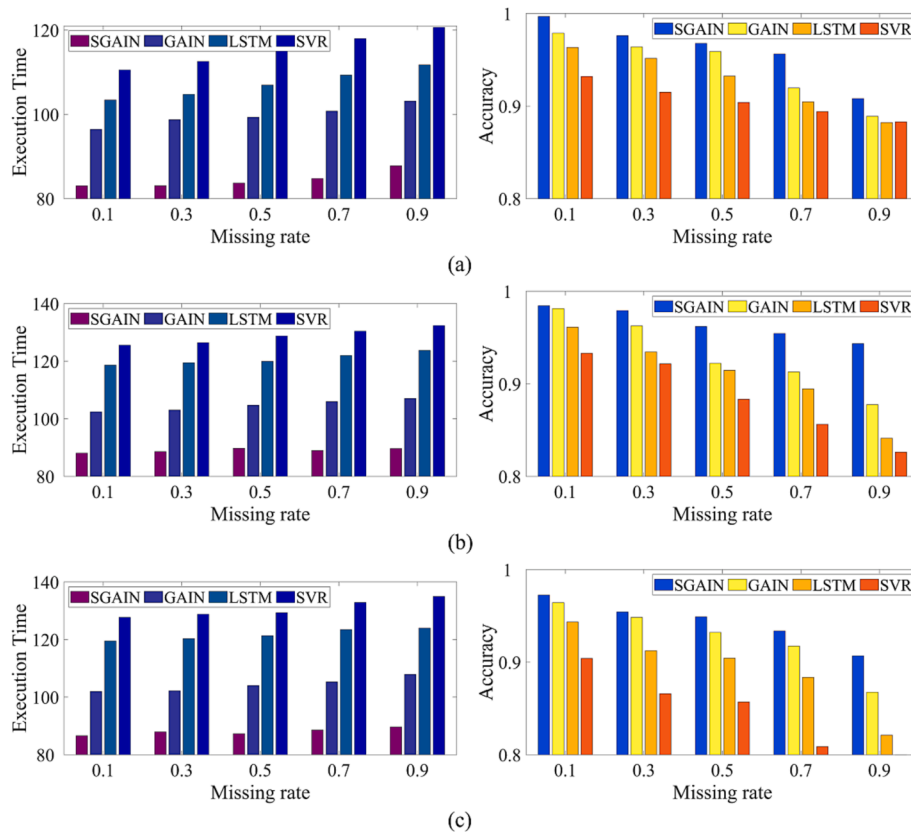


Fig. 19. Imputation capability assessment of SGAIN and the state-of-the-art imputation methods for deflection data (a) Type I; (b) Type II; (c) Type III.

Table 6

Imputation capability assessment of SGAIN and GAIN for deflection data.

Type	Ratio	SGAIN		GAIN		LSTM		SVR	
		ACC	Time	ACC	Time	ACC	Time	ACC	Time
Type I	10 %	0.9968	83.05	0.9788	96.439	0.9634	103.4123	0.932	110.532
	30 %	0.9761	83.09	0.9639	98.7231	0.9516	104.713	0.915	112.563
	50 %	0.9678	83.69	0.9589	99.33	0.9324	106.95	0.904	115.456
	70 %	0.9563	84.76	0.9195	100.77	0.9046	109.31	0.894	117.964
	90 %	0.9081	87.78	0.889	103.14	0.8823	111.74	0.883	120.64
Type II	10 %	0.9813	88.09	0.9843	102.40	0.9612	118.63	0.933	125.56
	30 %	0.9792	88.58	0.9626	103.03	0.9345	119.45	0.9216	126.45
	50 %	0.9619	89.79	0.9222	104.72	0.9146	120.014	0.8834	128.75
	70 %	0.9544	89.02	0.9129	105.96	0.8945	121.95	0.8563	130.43
	90 %	0.9434	89.65	0.8775	107.08	0.8412	123.72	0.8262	132.36
Type III	10 %	0.9724	86.58	0.9634	102.01	0.9434	119.46	0.904	127.69
	30 %	0.9542	87.33	0.9484	102.17	0.9123	120.34	0.8659	128.76
	50 %	0.949	87.93	0.9321	104.03	0.9043	121.35	0.8569	129.29
	70 %	0.9337	88.62	0.9174	105.26	0.8836	123.49	0.8087	132.87
	90 %	0.9067	89.7	0.8645	107.88	0.8212	123.98	0.7664	134.94

generalization ability and achieves the best performance in two missing types even with high missing proportions.

4. Discussion

To better understand the capability of the SGAIN model, the SGAIN, GAIN, LSTM, and SVR models [33,34] are utilized to recover the 1-day missing data in three scenarios (e.g., Type I, Type II, Type III) with different missing ratios and their results are compared. The algorithms are executed at a same computer configured as described in Section 3. Fig. 19 shows the quantitative imputation performance analysis for both

SGAIN and the other three imputation models. The SGAIN models can recover the missing deflection data with high accuracy and stability for three scenarios, and the accuracy are all over 0.9 in the case of high missing rate. This is mainly because the generator and discriminator powered by SGAIN framework can extract the important valuable information from non-missing part of fault sensor or other normal sensors. In detail, for Type-I, Type-II, and Type-II scenarios, the SGAIN model shows the better imputation performance with higher accuracy, and imputation efficiency with shorter execution time.

Comparatively, the other three models show poor imputation performance for all three scenarios. To be more specific, it can be seen from

Fig. 19 that the other three models show the worse imputation performance with lower accuracy, and low imputation efficiency with longer execution time. Table 6 also shows that the execution time of SGAIN model outperform GAIN with improvement ranging from 10 % to 20 % in three type scenarios, and the execution time of SGAIN model outperform SVR with improvement ranging from 20 % to 30 % in three type scenarios. Moreover, the accuracy of SGAIN model also outperform the other three models. It means the SGAIN model is significantly efficient with the accurate results, and it is believed that SGAIN have the good imputation capability when transferred to other valuable scenarios.

5. Conclusion

In this paper, a slim imputation framework based on generative adversarial networks has been proposed to recover the missing deflection data for bridge SHM systems. The deep learning method mainly includes generator and discriminator, which a slim architecture is utilized to capture the valuable information from non-missing part of fault sensor and other input sensors. Data imputation with different continuous missing rates (ranging from 1 h to 7 h) are implemented on the SGAIN framework to assess the imputation performance of different dataset correlation under three level input scenarios. Further, the imputation performance and efficiency of the SGAIN model are demonstrated for the three scenarios, namely single-sensor random loss without considering spatiotemporal correlation (one-to-one) as type I, single-sensor random loss considering spatiotemporal correlation (many-to-one) as type II, and single-sensor continuous loss considering spatiotemporal correlation (many-to-one) as type III. Six performance metrics including R2, RMSE, MAE, MNLL, Error, and Accuracy are utilized to assess the imputation accuracy and robustness. In addition, such three scenarios are implemented on SGAIN model and three state-of-the-art model comparatively, the accuracy and execution time are used to evaluate the imputation accuracy and efficiency.

Main conclusions are as follows:

1. The SGAIN model has outstanding imputation results under three-level correlation scenarios. In the long-term monitoring data, the mix dataset with various correlation between sensors is common. The imputation result of mixed correlation dataset (Level-III) is less accurate than better correlation datasets (Level-I and Level-II) for limited authentic information in the mixture features of Level-III.
2. The proposed method has a strong generalization ability and achieves the best performance in continuous and random missing types and high missing proportions. The fastest imputation error accumulation speed of random missing and continuous missing is in indifferent interval.
3. The proposed SGAIN model outperforms the conventional GAIN model for the three scenarios with different missing rates. The execution time of SGAIN model also outperforms GAIN with the improvement ranging from 10 % to 20 % in all these three scenarios and the execution time of SGAIN model outperform SVR with improvement ranging from 20 % to 30 % in three type scenarios. This improvement is significantly efficient with the accurate results, especially for high data missing rate.

It is believed that SGAIN could also have good imputation capability when transferred to other valuable scenarios. Further researches are required to investigate simultaneous missing data of the multiple sensors and improve the mode collapse and mode descend problem for response imputation using deep learning techniques.

CRedit authorship contribution statement

Shuai Gao: Methodology, Software, Writing – original draft. **Wenlong Zhao:** Validation. **Chunfeng Wan:** Writing – review & editing,

Supervision, Conceptualization, Resources. **Huachen Jiang:** . **Youliang Ding:** Funding acquisition. **Songtao Xue:** Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors are unable or have chosen not to specify which data has been used.

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